

FUTURE-MOBILITY-AWARE EV DECISION SYSTEM

Research Project Documentation

Consumer EV + Electric Vehicle Fleet

Literature Review • Research Gap • Problem Identification • Proposed Solution • Architecture • Workflow • Development Plan • Experiments • Benefits • Publication & Patent Strategy

Evidence focus: 2024–2026, with priority on 2026 research.

Prepared: 6 September 2026

Important: This is a research design document, not a guarantee of novelty, patentability, IEEE/Scopus acceptance, or commercial performance. All bibliographic and patent status should be re-checked before submission.

1. Executive Summary

This project proposes a common decision framework for two kinds of electric-vehicle users: an individual consumer who owns an EV and an organization operating many EVs as a fleet. The project starts from a simple observation: most EV routing and charging decisions focus strongly on the current trip or current operating period. Modern research already incorporates battery state, degradation, charging, traffic and uncertainty, so those features alone are not a sufficient research contribution.

The proposed research direction is to investigate **Future Mobility Feasibility**: the ability of an EV, or an EV fleet, to satisfy important future mobility requirements after making a decision today. The system estimates how today's route, charging or vehicle-assignment decision changes the future battery state and then estimates the probability that future mobility requirements will become infeasible.

For a consumer, future mobility means planned or expected upcoming trips. For a fleet, it means future customer, delivery, taxi or service demand. The common decision loop is: current state → candidate decision → battery/energy consequence → future state → future mobility requirement → future feasibility risk → current decision selection → execution → state update → re-optimization.

Research position: recent literature shows that battery-aware routing, aging-aware fleet routing, uncertain battery routing, personalized routing, multi-period planning and long-term battery management are already active areas. Therefore, this document deliberately does not claim that these individual components are novel. The candidate contribution is the explicit formulation and optimization of **future mobility feasibility risk** caused by current EV decisions under battery-health uncertainty, demonstrated consistently for both consumer and fleet settings. This must still be validated with a claim-level prior-art search before a patent claim is drafted.

2. Project in Simple English

What is the problem?

Imagine an EV owner asks: "Which route should I take today?" A normal system may choose the fastest route, cheapest route, or a route that has enough charging. But the owner may also have an important trip tomorrow. A decision that is good today can make tomorrow harder because of the battery state left after today's trip.

The same issue becomes larger for fleets. A fleet manager may assign the cheapest vehicle to a long route today, but that vehicle may then have less useful capacity or require charging/maintenance when demand is high tomorrow.

What is the proposed idea?

The system asks two questions together: **Can I satisfy today's requirement?** and **Will today's decision leave enough capability to satisfy future requirements?**

Consumer example

- Today: 150 km journey; tomorrow: another important 180 km journey.
- Route A is fastest but creates higher predicted battery stress.
- Route B is slightly slower but leaves a better predicted state for tomorrow.
- The system can choose Route B if the improvement in future feasibility is worth the time/cost trade-off.

Fleet example

- Today there are 20 delivery tasks and tomorrow demand is predicted to be high.
- The optimizer avoids using the same vulnerable vehicles heavily today when doing so would increase tomorrow's service-failure risk.
- The result is not simply "use the healthiest battery"; it is a decision that balances today's service with future fleet capability.

3. 5Ws + 1H

Question	Project answer
What?	A future-mobility-feasibility decision framework for EV routing, charging and fleet assignment under battery-health and future-demand uncertainty.
Why?	Because current decisions can change future battery capability, future range and future fleet availability. Existing systems often optimize current performance or specific long-term objectives, but the proposed work makes future mobility feasibility/risk an explicit decision variable.
Who?	Consumer EV owners, fleet managers, delivery/taxi operators, OEM mobility services, charging operators and researchers.
Where?	Consumer navigation/energy-management software and fleet planning/dispatch systems; first implemented in simulation, then optionally HIL.
When?	During route planning, before charging, during changing conditions, and repeatedly across planning periods as battery state and demand forecasts change.
How?	Predict energy use and battery consequence for candidate actions, forecast future battery capability, model future trips/tasks, calculate future feasibility risk, optimize current decisions, execute, observe, and re-optimize.

4. Research Objectives

- Define a measurable Future Mobility Feasibility (FMF) metric for a consumer EV and an EV fleet.
- Define Future Mobility Risk (FMR) as the probability or expected severity of failing a future mobility requirement.
- Build a battery/energy state transition model linking today's routing/charging decisions to future state.
- Develop a decision optimizer that includes current operational objectives and future-feasibility constraints.
- Support uncertainty in battery state, energy consumption, traffic/temperature and future demand.
- Compare the proposed framework with established routing and degradation-aware baselines.
- Demonstrate both consumer and fleet use cases without pretending that the two have identical decision variables.
- Quantify trade-offs among time, money, energy, battery degradation and future mobility reliability.

5. Research Questions

- RQ1: How should future mobility feasibility be defined for an individual EV and an EV fleet?
- RQ2: How strongly do current routing and charging choices change future mobility feasibility?
- RQ3: Can uncertainty in battery state and future demand be incorporated without making the optimization impractical?
- RQ4: Does explicitly minimizing future mobility risk improve future service feasibility compared with current-trip optimization?
- RQ5: What is the trade-off between future feasibility protection and current travel time/cost?
- RQ6: Does the framework remain useful under different battery-health levels, demand scenarios and charging conditions?

6. Recent Literature Review: 2026 Priority

The following sources are particularly important because they show what is already solved and therefore prevent the project from claiming weak novelty.

Paper	Year/Venue	What it does	Limitation relevant to this project
Real-time electric vehicle routing and charging optimization with battery life consideration	2026, Internet of Things / Elsevier	Integrates traffic, charging availability, SoC constraints and battery-life consideration; compares heuristics, Q-learning, DQN, simulated annealing and memetic methods.	Shows that real-time routing + charging + battery life is already an active and technically mature direction. A new paper must contribute more than these ingredients.
Aging-aware fleet management for electric vehicle routing problem	2026, Computers & Industrial Engineering	Uses a real-time/reconfigurable battery model with SoC/SoH and multi-objective optimization balancing energy, battery lifespan and quality of service.	Directly overlaps generic aging-aware fleet optimization. Future-feasibility formulation must be more specific.
Mitigate the range anxiety: two-stage optimization for EV routing with time windows and battery status uncertainty	2026 issue / 2025 DOI, Transportation Research Part E	Models uncertain battery status using stochastic scenarios and two-stage optimization with recourse.	Shows that battery uncertainty + robust/stochastic routing is already established.
Battery-Degradation-Aware Routing to Nearest Feasible Charging Station for Electric Vehicles	2026, World Electric Vehicle Journal	Uses real-time battery state estimation including temperature, SoC and mechanical degradation/swelling to select feasible charging stations.	Shows that high-fidelity degradation-aware consumer routing is already being explored.
Adaptive electric vehicle routing and charging with deep reinforcement learning	2026, Transportation Research Interdisciplinary Perspectives	Combines travel-time uncertainty learning, time-dependent shortest paths and DQN for adaptive charging and routing; supports multiple user preferences.	Shows personalized, uncertainty-aware real-time routing/charging is already covered.

These papers collectively mean that “battery-aware routing”, “battery degradation”, “charging optimization”, “uncertainty”, “RL” and “personalization” should be treated as existing building blocks rather than claimed as the invention.

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7. 2025 and 2024 Literature

Paper	Year/Venue	Main contribution	Why it matters
Multi-objective optimization model for routing flexibility and long-term EV sustainability	2025, Procedia Computer Science	Long-term fleet task allocation, battery degradation and maintenance-battery operations.	Shows that long-term fleet effects of task allocation are already studied.
A review of static and dynamic charging in electric vehicle routing: Transition, algorithms and future directions	2025, Swarm and Evolutionary Computation	Systematic review of 140 EVRP studies, dynamic charging and algorithmic approaches.	Useful survey confirming how crowded EVRP is and identifying future directions.
Joint electric vehicle routing and battery health management integrating an explicit state of charge model	2024, Computers & Industrial Engineering	EVRP with realistic SoC/SoH, degradation, speed/acceleration limitations and genetic programming.	Direct evidence that routing + realistic battery health is already established.
Electric vehicle scheduling: State of the art, critical challenges, and future research opportunities	2024, Journal of Industrial Information Integration	Structured survey covering EV scheduling, charging, grid/environment considerations and future opportunities.	Useful background for defining the research space.
Towards an intelligent battery management system for EV applications	2024, Journal of Energy Storage	Reviews SOH estimation methods and open battery-health data/model challenges.	Important for choosing the battery-health prediction component.

The 2024–2026 literature shows a progression: realistic battery models → degradation-aware routing → uncertain battery routing → real-time routing/charging → long-term fleet sustainability. The proposed project therefore needs a precise contribution beyond adding another battery variable.

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8. Literature Survey: What Is Already Solved?

Area	Current state	Project implication
Shortest/fastest EV routing	Mature	Not novel.
Energy-aware routing	Mature	Not novel.
Charging-station selection	Mature/active	Not novel.
SoC-aware routing	Mature	Not novel.
SoH/battery degradation-aware routing	Strong 2024–2026 literature	Not sufficient alone.
Battery-status uncertainty	Recent stochastic routing literature	Not sufficient alone.
Real-time traffic-aware routing	Strong active area	Not sufficient alone.
Personalized routing/charging	2026 work exists	Not sufficient alone.
Aging-aware fleet routing	2026 direct literature	Not sufficient alone.
Long-term fleet task allocation	2025 direct literature	Not sufficient alone.
Future-demand-aware fleet planning	Existing literature/patent family	Not sufficient alone.
Future mobility feasibility as an explicit probabilistic constraint linked to current battery-health decisions	Candidate area	Needs dedicated search and validation.

9. Literature Limitations

- Many routing studies optimize current operational cost, travel time, energy or service quality.
- Battery-health models can be simplified or may be accurate but computationally expensive.
- Uncertainty is often handled for one component—such as battery status, travel time or demand—rather than as a unified current-to-future feasibility problem.
- Long-term fleet optimization exists, so the research must avoid claiming that considering multiple periods is new.
- Consumer future-trip planning also exists in the broader literature/patent landscape; the contribution must be a specific mathematical mechanism rather than simply storing future trips.
- Dynamic charging and RL are active research areas; using DQN or similar methods is not itself a novelty claim.
- Large-scale optimization may become computationally difficult when battery health, routing, charging, future demand and uncertainty are combined.

10. Candidate Research Gap

Candidate gap: A decision framework that explicitly converts the consequence of a current EV routing/charging/assignment decision into a measurable probability of future mobility infeasibility, while accounting for battery-health uncertainty and future mobility requirements, and then uses that risk as an explicit constraint or objective for the current decision.

This is deliberately narrower than “future-aware EV routing.” The proposed scientific object is **future feasibility**: whether future requirements remain satisfiable after the present decision.

Why this is different

Existing approach	Candidate approach
Minimize today's travel/energy cost	Optimize today's cost subject to future feasibility.
Minimize battery degradation	Minimize degradation while protecting future mobility requirements.
Predict battery health	Use predicted battery state to evaluate future mobility feasibility.
Forecast demand	Use demand forecast to define future feasibility requirements.
Multi-period planning	Explicitly quantify future infeasibility risk caused by current decisions.
Generic risk-aware routing	Tie risk to future mobility requirements and battery-state transition.

Novelty warning: this is a candidate research gap, not a confirmed patent novelty conclusion. The final stage must include a patent-family and claim-element search specifically for future-trip feasibility, battery-health constrained planning, future-demand capacity and current-to-future decision coupling.

11. Exact Problem Statement

Problem: Current EV routing and fleet-management systems can optimize route, energy, charging, battery degradation and uncertainty, but there remains a research opportunity to formulate the future mobility consequence of a current decision as an explicit feasibility/risk measure. Without this link, a decision that is locally optimal may reduce the vehicle's or fleet's ability to satisfy an important future mobility requirement.

Research problem: Develop and evaluate a decision framework that estimates the future mobility feasibility of an EV or EV fleet after a candidate current routing/charging/assignment action under uncertain battery health and future demand, and selects the current action that balances present operational performance with future feasibility.

12. Proposed Solution

The solution has five major engines.

- **1. Current-state engine:** captures SoC, SoH, temperature, battery capacity, vehicle efficiency, current location and operational state.
- **2. Consequence engine:** predicts energy use and battery-state change for each candidate route, charging action or vehicle assignment.
- **3. Future-state engine:** predicts a distribution/range of future SoC/SoH/available energy rather than one perfect value.
- **4. Future-demand engine:** represents future consumer trips or fleet task demand as known, probabilistic or scenario-based requirements.
- **5. Decision engine:** chooses the current action using current cost plus future feasibility risk/constraint.

13. Core Concepts

13.1 Future Mobility Feasibility — FMF

FMF measures whether the vehicle or fleet can satisfy the defined future mobility requirements under the modeled state and uncertainty.

Consumer example: probability that all planned trips in the next planning horizon can be completed, including required charging stops.

Fleet example: probability that the fleet can cover the required future tasks within service windows and resource constraints.

13.2 Future Mobility Risk — FMR

A simple conceptual definition is:

FMR = Probability(future mobility requirements become infeasible)

The final implementation can use Monte Carlo simulation, scenario optimization, chance constraints or another mathematically justified method.

13.3 Future Mobility Capacity

Future Mobility Capacity is the remaining ability of a vehicle/fleet to meet expected future mobility requirements after the current decision. It should be derived from measurable quantities rather than an arbitrary health score.

14. Consumer EV Architecture

The consumer architecture is designed around a personal trip planner.

Module	Function
User profile	Future trips, time preferences, charging preferences, battery-care preference.
Vehicle state	SoC, SoH, temperature, battery capacity, current location, efficiency.
Energy predictor	Predict route energy consumption from distance, speed, traffic, gradient, weather and vehicle state.
Battery consequence model	Estimate future battery state/degradation under route and charging candidates.
Charging engine	Evaluate station location, power, price, waiting time and charging duration.
Future-trip engine	Represent planned future journeys and their requirements.
FMF/FMR engine	Calculate future trip feasibility/risk after each candidate current action.
Optimizer	Choose route + charging strategy subject to user and future-feasibility preferences.
Explanation engine	Explain why the selected option protects future mobility.
Feedback loop	Update estimates using actual trip and charging results.

15. Fleet Architecture

Module	Function
Fleet state	SoC/SoH/temperature/location/status for every vehicle.
Task manager	Current jobs, time windows, priorities and constraints.
Demand forecast	Expected future jobs/tasks with uncertainty scenarios.
Battery consequence model	Estimate health/energy effect of assignments and routes.
Charging planner	Plan current/future charging and queue constraints.
Future capacity engine	Estimate whether fleet can satisfy future demand.
Risk engine	Estimate future service infeasibility probability.
Fleet optimizer	Jointly select assignment, routes and charging.
Dispatcher	Send current decisions to operations.
Learning loop	Update models from actual vehicle/task/battery outcomes.

16. Unified Architecture

The same scientific core supports both applications, while consumer and fleet decision layers use different variables.

Current state → Candidate action generator → Energy prediction → Battery state transition → Future state distribution → Future mobility requirement → Feasibility simulation → FMR/FMF calculation → Multi-objective/chance-constrained optimizer → Decision explanation → Execution → Telemetry → Model update → Re-optimization.

Conceptual architecture:

Data Sources → State Estimation → Energy Prediction → Battery Consequence Model → Future Mobility Model → Future Risk Engine → Optimizer → Recommendation/Dispatch → Execution → Feedback.

Key data objects

Object	Important fields
VehicleState	vehicle_id, time, SoC, SoH, temperature, capacity, location, efficiency
RouteCandidate	route_id, distance, time, traffic, gradient, energy
ChargeCandidate	station, arrival SoC, power, price, wait, duration
BatteryTransition	initial state, action, predicted next state, uncertainty
FutureTrip	destination, distance, deadline, required SoC/charging
FutureTask	fleet customer/job, time window, service requirement
Scenario	traffic, weather, demand, battery realization
FeasibilityResult	scenario, feasible/infeasible, reason
Decision	chosen action, objective, risk, constraints
Outcome	actual energy, SoC, time, charging, deviations

17. End-to-End Workflow

1. Collect current vehicle/fleet state.
2. Collect current trip/task requirements.
3. Collect future consumer trips or forecast future fleet demand.
4. Generate candidate routes, charging plans and, for fleets, vehicle assignments.
5. Predict energy consumption for each candidate.
6. Predict battery-state consequence and uncertainty.
7. Project the future vehicle/fleet state after the candidate decision.
8. Generate future demand scenarios.
9. Test whether future requirements can be satisfied under each scenario.
10. Calculate Future Mobility Feasibility and Future Mobility Risk.
11. Optimize current cost, energy, battery consequence and future risk.
12. Return the best decision and an explanation.
13. Execute the route/assignment/charging plan.
14. Measure actual outcome.
15. Update battery/energy/demand models.
16. Repeat when the state changes materially.

18. Mathematical Formulation — Initial Version

Let a current decision be a. Let x_t represent current vehicle/fleet state and ω represent future uncertainty.

State transition: $x_{t+1} = F(x_t, a, \omega)$.

Future feasibility: $I_{\text{future}}(a, \omega) = 1$ if all required future trips/tasks can be completed; otherwise 0.

Future risk: $FMR(a) = P[I_{\text{future}}(a, \omega) = 0]$.

A conceptual optimization is:

$\min J(a) = \text{CurrentCost}(a) + \lambda \cdot \text{BatteryConsequence}(a) + \mu \cdot FMR(a)$

subject to route, SoC, charging, time-window, vehicle-capacity and safety constraints.

Alternatively, future risk can be a hard chance constraint:

$FMR(a) \leq \varepsilon$

where ε is an allowed future infeasibility probability selected for the experiment.

Important

Do not choose λ , μ or ε arbitrarily and then claim they are universally correct. The study should run sensitivity analysis and explain how different user/fleet priorities change the decision.

19. Development from Zero

Phase	Development	Deliverable
1. Scope	Freeze consumer + fleet scenarios, planning horizon and assumptions.	Research specification.
2. Literature	Build 2024–2026 paper matrix and patent matrix.	Gap evidence.
3. Data model	Define vehicle, route, charge, future trip/task and scenario schemas.	Versioned schemas.
4. Simulator	Build road/traffic/charger/vehicle simulation.	Reproducible environment.
5. Energy model	Implement route-energy prediction.	Energy module.
6. Battery model	Implement SoC/SoH/degradation/state transition.	Battery module.
7. Future demand	Consumer planned trips; fleet demand scenarios.	Future-demand module.
8. Feasibility engine	Simulate future requirements after candidate decisions.	FMF/FMR module.
9. Optimizer	Implement baseline and proposed optimization.	Decision engine.
10. Explanation	Generate human-readable reasons for decisions.	Explainability layer.
11. Experiments	Run scenarios, baselines and ablations.	Results dataset.
12. Validation	Sensitivity, robustness, runtime overhead and reproducibility.	Evaluation report.
13. Prototype UI	Consumer app-like screen and fleet dashboard.	Demonstration system.
14. Paper/patent	Write paper; conduct final claim search.	Submission package.

20. Recommended Technology Stack

Layer	Initial choice	Purpose
Programming	Python	Fast research iteration.
Optimization	Pyomo + MILP solver; OR-Tools where appropriate	Optimization baselines.
Graph routing	NetworkX / custom A*	Route generation.
Battery model	Python numerical model	SoC/SoH/degradation simulation.
ML	scikit-learn first	Energy/demand prediction.
Advanced ML	PyTorch only if needed	Compare advanced prediction if justified.
Simulation	SUMO and/or CARLA	Traffic/vehicle environment.
Data	Pandas + Parquet/SQLite	Experiment storage.
Dashboard	Streamlit initially	Fast prototype.
API	FastAPI optional	Service layer.
Reproducibility	Git + Docker	Repeatable experiments.

21. Experimental Design

Consumer scenarios

- Normal trip with no future conflict.
- Long future trip tomorrow.
- Low SoH vehicle.
- High-temperature day.
- Heavy traffic.
- Charger unavailable after route selection.
- High charging price at the closest station.
- Uncertain energy consumption.
- Unexpected future trip.
- Different user preferences: fastest, cheapest, battery-care, future-ready.

Fleet scenarios

- Normal demand.
- High demand tomorrow.
- Uneven battery health across vehicles.
- Limited charging capacity.
- Multiple vehicles competing for chargers.
- Vehicle temporarily unavailable.
- Demand forecast error.
- Battery-state prediction error.
- High traffic/energy conditions.
- Sudden priority task.

Baselines

- Shortest/fastest route.
- Energy-aware route.
- Battery-degradation-aware route.
- Battery-uncertainty-aware routing.
- Long-term/multi-period baseline where available.
- Proposed FMF/FMR-aware method.

22. Evaluation Metrics

Metric	Meaning
Current travel time	Immediate user/operation performance.
Energy consumption	Immediate efficiency.
Charging cost	Money spent on charging.
Charging time	Time spent charging.
Battery degradation proxy	Estimated health consequence.
Future trip feasibility	Fraction of future consumer scenarios completed.
Future fleet task completion	Fraction of future tasks successfully served.
Future mobility risk	Probability of future infeasibility.
Service failure	Missed/late future tasks in fleet.
Computation time	Practicality of the optimizer.
Robustness	Performance under forecast/model errors.
User preference satisfaction	Consumer trade-off quality.

Ablation studies

- Remove future-feasibility term.
- Remove battery-health uncertainty.
- Remove future-demand uncertainty.
- Remove charging uncertainty.
- Remove battery degradation.
- Remove user preferences.
- Use deterministic instead of probabilistic feasibility.
- Change planning horizon.

The ablation results are important because they show which component actually creates the improvement. Without ablation, a reviewer can reasonably argue that the proposed framework is only a larger collection of existing features.

23. Real Problems and Why the Project Could Help

The research is motivated by real operational problems: EV drivers experience range anxiety and uncertainty about charging; battery aging changes usable capability; fleet operators must handle limited battery capacity, charging and vehicle availability; and routing decisions can affect long-term battery use.

Recent research itself describes range anxiety and battery-status uncertainty as practical routing problems. The 2026 battery-degradation-aware routing paper specifically discusses thermal and mechanical battery conditions, while the 2026 stochastic EV routing paper addresses uncertainty that can make planned routes infeasible.

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Important real-world interpretation

The project is not claiming that one published accident was caused by a missing future-mobility optimizer. That would be unsupported. Instead, the real problem is an operational class: a vehicle can be able to complete today's trip while future range, charging access, battery condition or fleet availability becomes insufficient for later requirements.

Potential consumer benefits

- Lower probability of future-trip failure.
- Better coordination between today's route and tomorrow's planned trip.
- Potential reduction in unnecessary charging or high-stress charging decisions.
- Clearer trade-off between travel time, cost and battery preservation.
- More explainable route/charging recommendations.

Potential fleet benefits

- Lower probability of future service failures.
- Better allocation of heterogeneous vehicle health.
- Potentially lower battery-related operating cost.
- Better use of charging capacity.
- Improved planning under demand uncertainty.
- Better visibility into how today's decisions affect future fleet capability.

24. Money and Business Benefit Model

Do not invent a percentage of savings. The research should calculate benefits from measured experiments or transparent cost assumptions.

A simple consumer model can be:

Total Cost = travel cost + charging cost + estimated battery degradation cost + expected future disruption cost.

A fleet model can be:

Total Cost = energy + charging + delay/service penalty + battery degradation/maintenance + expected future service-failure cost.

Expected future disruption cost can be modeled as:

Expected Cost = Probability of failure × Cost of failure.

For a fleet, cost of failure may include an extra vehicle, late delivery penalty, overtime, emergency charging, or lost service. These must be based on a documented assumption or real operational data.

25. End Product — What Will You Actually Build?

The final product should be a **research prototype**, not just a paper or a dashboard.

Part	Final output
Consumer planner	Web/app-like interface showing current route, charging options and future-mobility risk.
Fleet planner	Dashboard showing vehicle health, assignments, future demand and future-capacity risk.
Decision engine	Routing + charging + assignment optimizer.
Battery engine	SoC/SoH/degradation/state-transition simulator.
Future feasibility engine	Future trip/task scenario evaluator.
Prediction engine	Energy and demand forecasting.
Experiment engine	Automated scenario generation and evaluation.
Results dashboard	Metrics, comparisons, risk curves and trade-off plots.
Research dataset	Generated/processed experiment records.
Codebase	Reproducible implementation with configuration files.
Paper	Full scientific manuscript.
Patent package	Only if final prior-art search supports a defensible technical claim.

26. What the User Sees

Consumer

The consumer sees three or more route/charging choices with travel time, cost, battery impact and future-trip feasibility. The system explains why a recommendation was selected.

Fleet manager

The fleet manager sees current fleet health, current tasks, future demand, predicted capacity risk and recommended assignments. The system can identify which vehicle should be protected or charged and why.

27. Publication Strategy

The project has a plausible publication path if the final research produces a specific mathematical contribution and strong experimental evidence. A Google Scholar appearance is normally a consequence of legitimate publication/indexing and is not itself a quality guarantee. Scopus and IEEE acceptance depend on venue-specific standards.

Paper contribution structure

- Problem and motivation.
- 2024–2026 literature review.
- Precise gap definition.
- Future Mobility Feasibility/Future Mobility Risk formalization.
- Battery-health and uncertainty model.
- Optimization formulation.
- Algorithm.
- Consumer implementation.
- Fleet implementation.
- Experimental benchmark.
- Baseline comparison.
- Ablation and sensitivity analysis.
- Limitations and threats to validity.
- Conclusion.

What reviewers will challenge

- Is this really different from multi-period EV routing?
- Is future demand already handled by existing fleet planning?
- Is future-trip feasibility already present in commercial/patent systems?
- Why is a probabilistic risk formulation technically useful?
- How accurate is the battery degradation model?
- How much computation is required?
- Does the method improve outcomes enough to justify complexity?
- Are consumer and fleet experiments truly connected by one framework?

28. Patent Strategy

Patent potential is more uncertain than publication potential because patentability depends on novelty, inventive step/non-obviousness, claim wording and jurisdiction.

Do not claim

- EV routing in general.
- Battery-aware routing.
- Battery degradation-aware routing.
- Charging recommendation.
- Future trip planning alone.
- Fleet task allocation alone.
- Demand forecasting alone.
- AI/ML for EV routing.

Candidate technical claim direction

Investigate whether a defensible claim can be built around a specific machine-implemented mechanism that: (1) receives current battery state and a candidate decision, (2) predicts the resulting future state distribution, (3) represents future mobility requirements, (4) computes future infeasibility probability, and (5) modifies the current routing/charging/assignment decision to satisfy a defined future-feasibility constraint.

This must be tested against patent families before filing. The document should not be used as a patentability opinion.

29. Limitations of the Proposed Research

- Battery degradation is difficult to model accurately and is path-dependent.
- Future consumer trips may be unknown or change unexpectedly.
- Fleet demand forecasts can be wrong.
- Traffic and charger availability are uncertain.
- Optimization complexity can grow rapidly.
- Simulation results may not represent every real EV.
- Future mobility risk depends on the chosen planning horizon.
- Cost models for battery degradation and service failure contain assumptions.
- Consumer preferences may not be measurable precisely.
- A research prototype is not a safety-certified automotive product.

30. Threats to Validity

Threat	Mitigation
Battery-model mismatch	Use multiple battery models/sensitivity ranges; compare with public datasets where possible.
Synthetic future demand	Use several distributions and, if possible, historical demand data.
Traffic realism	Use real network traces or established traffic simulators.
Optimizer bias	Use multiple baselines and independent implementations where practical.
Hyperparameter bias	Predefine tuning protocol and report sensitivity.
Small scenarios	Test both small exact cases and larger heuristic cases.
Overclaiming	Separate measured results from assumptions and projections.

31. Project Timeline

Stage	Approx. duration	Main output
Literature + patent search	2–3 weeks	Final gap and claim chart.
Problem/model definition	1–2 weeks	Formal model.
Data/simulation setup	2–3 weeks	Reproducible environment.
Battery/energy model	2–4 weeks	State transition module.
Future feasibility engine	2–3 weeks	FMF/FMR engine.
Consumer optimizer	2–3 weeks	Consumer prototype.
Fleet optimizer	3–4 weeks	Fleet prototype.
Baselines	2 weeks	Comparison systems.
Experiments	3–5 weeks	Results.
Ablation/sensitivity	2 weeks	Scientific evidence.
Dashboard	1–2 weeks	Demonstration.
Paper	3–4 weeks	Manuscript.
Patent decision	1–2 weeks	Invention disclosure/claim chart if justified.

32. Minimum Viable Research Prototype

If time is limited, do not build a huge production application. Build one strong scientific prototype.

- One realistic consumer scenario.
- One realistic fleet scenario.
- One battery-health model.
- One future-demand model.
- One Future Mobility Feasibility metric.
- One Future Mobility Risk calculation.
- One optimizer.
- Three strong baselines.
- At least 8–10 controlled scenarios.
- Repeated experiments with statistical analysis.
- Clear plots showing current cost versus future feasibility.

33. Final Research Position

Recommended project name: Future-Mobility-Aware EV Decision System

Scientific focus: Future Mobility Feasibility under Battery Health and Demand Uncertainty.

Core question: How should an EV or EV fleet make today's routing, charging and assignment decisions when those decisions can change the ability to satisfy future mobility requirements?

Why this is stronger than the original fleet-only idea

It does not depend on one narrow commercial application. The same scientific concept is evaluated for an individual EV and a fleet. It also avoids claiming that battery-aware routing itself is new, because recent 2024–2026 research already covers that area extensively.

Keep / Modify / Drop decision

Candidate	Decision
Generic battery-aware EV routing	DROP as a novelty claim.
Battery-degradation-aware fleet routing	DROP as a novelty claim.
Future-demand-aware fleet routing	MODIFY; substantial prior work exists.
Consumer future-trip-aware routing	MODIFY; patent/research overlap must be checked.
Future Mobility Feasibility + current decision + battery-state transition + uncertainty	KEEP as the main candidate, subject to final prior-art search.
Unified consumer + fleet validation	KEEP as a research-strengthening contribution.

Final caution

The strongest evidence available now supports this as a **research candidate**, not a guaranteed novel invention. Before coding the full system, conduct a focused 2026/2025/2024 patent and literature search for the exact chain: current decision → battery-state transition → future-trip/task feasibility → infeasibility probability → current decision modification.

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35. Final One-Page Explanation

The project is about making an EV smarter about the **future**. Current EV navigation can already find routes, charging stations and energy-efficient paths. Research can also consider battery degradation, battery uncertainty, traffic and fleet planning. Therefore, the project does not claim those basic features as new.

Instead, the proposed system asks: **"After I make this decision today, can my EV or fleet still satisfy the mobility requirements I expect in the future?"**

For a consumer, the future requirement may be tomorrow's trip. For a fleet, it may be tomorrow's delivery/taxi/service demand. The system predicts the battery consequence of candidate decisions, models uncertainty, checks future feasibility, calculates future mobility risk, and uses that risk to choose today's route, charging strategy or vehicle assignment.

The final product is a research prototype with a consumer planner, fleet planner, battery/energy model, future-feasibility engine, optimizer, explanation layer, simulator and evaluation framework. The scientific output is the mathematical definition, algorithm and experimental evidence. The commercial output could eventually be a navigation/fleet decision service, but commercialization is outside the first research scope.

Most important: do not start by claiming "this is novel." Start by proving the exact gap. If the final literature/patent search shows the exact decision chain is already covered, narrow the contribution. If it is not covered and the prototype demonstrates measurable benefit, the project becomes a much stronger candidate for an IEEE/Scopus paper and potentially a patent.

END OF FULL PROJECT DOCUMENT